Synology DS412+

RAM Upgrade Guide

Upgrade	1 GB → 4 GB DDR3
Time	~30 minutes
Difficulty	Moderate — requires full disassembly
Tools	Phillips PH1 screwdriver + anti-static precaution
Cost	~€15–25 for the RAM module
Warranty	Voids Synology warranty (device is end-of-life anyway)

What You Need Before Starting

Compatible RAM Modules (community-tested)

The DS412+ uses a single DDR3 SO-DIMM slot. The stock module is 1 GB. The Intel Atom D2700 officially supports up to 4 GB. Community reports confirm these modules work:

Module	Capacity	Speed	Notes
G.Skill F3-8500CL7S-4GBSQ	4 GB	DDR3-1066	✓ Best choice — confirmed by multiple users
Kingston KCP313SS8/4	4 GB	DDR3-1333	✓ Confirmed — now discontinued, may find NOS
Kingston KVR1333D3S9/4G	4 GB	DDR3-1333	✓ Confirmed working — passed 8h memory test
Samsung M471B5273DH0-CH9	4 GB	DDR3-1333	✓ Confirmed — community tested
Transcend TS512MSK64V3H	4 GB	DDR3-1333	✓ Compatible spec — use 1.5V version
Samsung M471B5773CHS-CH9	2 GB	DDR3-1333	✓ Confirmed working — if 4 GB causes issues
Kingston KVR13S9S6/2	2 GB	DDR3-1333	✓ Safe conservative option — no boot issues

■ 4 GB NOTE — Read before ordering

Some users report slowness or freezing with 4 GB. This is rare and module-dependent. The G.Skill F3-8500CL7S-4GBSQ and Kingston KVR1333D3S9/4G have the most positive reports. If you experience issues with 4 GB, drop to 2 GB — it runs reliably and still doubles the RAM for Docker.

Tools required

Phillips PH1 screwdriver	REQUIRED	All screws are Phillips head. PH1 tip fits perfectly — PH0 is too small, PH2 too large.
Anti-static wrist strap	STRONGLY RECOMMENDED	Or touch a grounded metal object (e.g. radiator) before touching the motherboard.
Compressed air can	RECOMMENDED	Your DS412+ has years of dust inside. While you're in there, clean it out.
Plastic spudger / spoon	HELPFUL	To help release the plastic side tabs without cracking them (see Step 4).
Flashlight / phone torch	HELPFUL	The inside is dark and the screw holes can be hard to see.
Small container for screws	HELPFUL	There are 7+ screws. Keep them organised — they are the same size.

1 Shut Down the NAS Properly

~5 min

DO NOT hot-swap the RAM

The DS412+ must be completely powered off before opening it. Do NOT just pull the power cable — use the DSM shutdown to let the drives park safely.

- In DSM web interface: click your username (top-right) → Shut Down
- Wait for the power LED to turn off completely (~2–3 minutes for drives to spin down)
- Unplug the power cable from the rear
- Unplug all ethernet cables
- Wait 30 seconds for capacitors to discharge
- Touch a grounded metal object to discharge static from your body

2 Remove the Hard Disk Trays

~3 min

This is necessary to access the screws that hold the outer shell and the internal cage. You do NOT need to disconnect the drives from their trays.

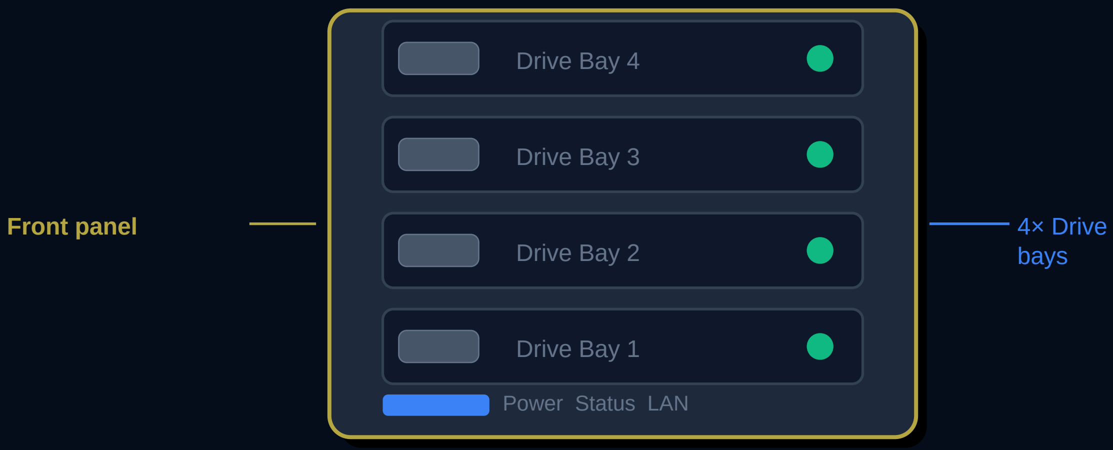

Figure 1: Synology DS412+ front view

- Pull the tray handle outward (it has a small latch — press the latch button first)
- Slide each tray out straight toward you
- Repeat for all 4 bays
- Set the drives aside on a flat, safe surface
- Number/label them if you want to put them back in the same order (good practice with RAID)

3 Remove the 2 Rear Panel Screws

~3 min

With the NAS on a flat surface and the rear panel facing you, locate and remove two Phillips screws — one near the top edge and one near the bottom edge of the rear.

Tool needed: Phillips PH1 screwdriver

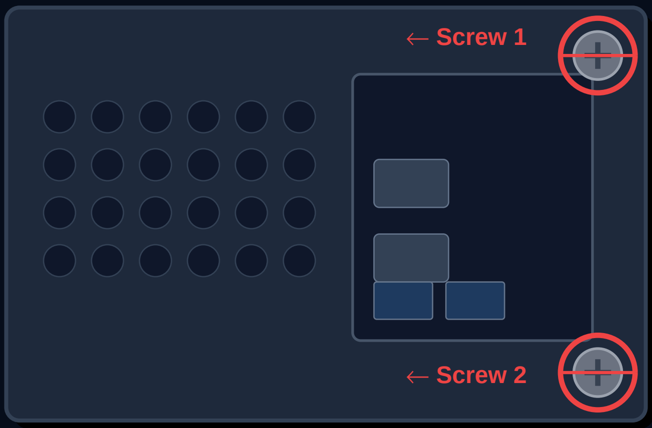

Figure 2: Rear panel — remove these 2 screws first

- Use your PH1 Phillips screwdriver
- Screw 1 is near the TOP of the rear panel (approximately centred left-right)
- Screw 2 is near the BOTTOM of the rear panel
- Both screws are the same size — keep them together
- These screws allow the outer shell to slide backward off the chassis

4 Release the Side Tabs (Trickiest Step)

~5 min

■ Most common damage point

The outer plastic shell is held by 2 plastic tabs on the LEFT side that hook over a metal bar. If you try to pry from the outside, you will crack the tabs. Reach inside from the front (where the drives were) and push the tabs UPWARD.

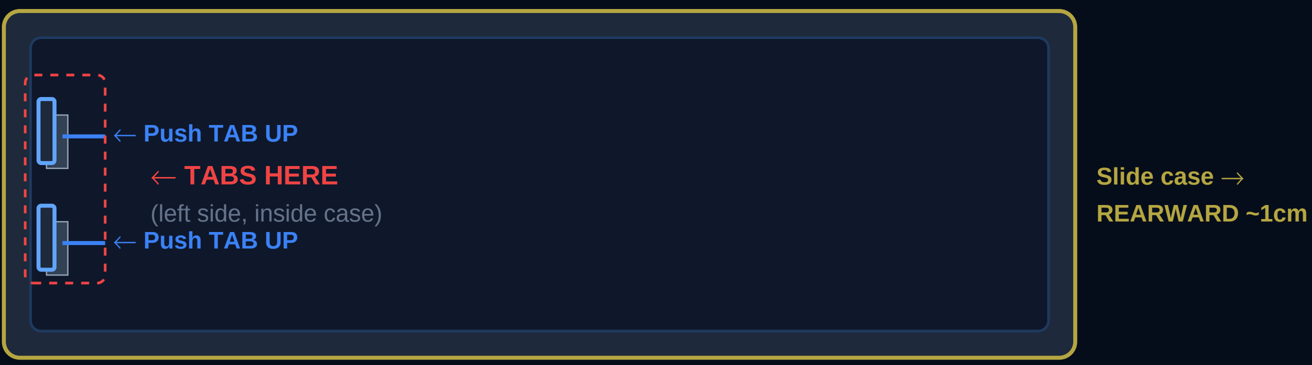

Figure 3: Left-side tabs (cross-section top view) — lift both tabs UP off the metal bars

- Place the NAS on its RIGHT side (so the left side faces up)
- Reach inside through the FRONT drive bay opening
- There are 2 plastic tabs — one near the front-top, one near the front-bottom of the left side
- Using your finger or a plastic spudger, push each tab UPWARD off the metal bar
- Tip from the iFixit community: use a small spoon as a lever from inside against the metal beam
- You should feel them release — do NOT force them if they feel very stiff (check you got both tabs)

5 Slide and Lift the Outer Shell Off

~3 min

Figure 4: Slide case ~1 cm toward rear, then lift straight up

- With both tabs released, the outer shell is free
- Slide the shell approximately 1 cm toward the REAR of the unit
- Then LIFT the shell straight UP — mind the tabs on the top edge which disengage as you lift
- The shell should come away cleanly in two motions: slide back, then lift

- If it resists, check that both tabs are fully released and the rear screws are out
- Set the outer shell aside

6 Clean the Interior (While You're In Here)

~5 min

Do not skip this

A DS412+ that has been running for years will have significant dust buildup on the fans, heatsink, and circuit boards. Dust causes thermal throttling, reduced lifespan, and fan noise. This is your best opportunity to clean it — takes 3 minutes.

Before proceeding with disassembly, take a moment to clean:

- Hold the compressed air can upright and spray short bursts
- Clean the fan blades (hold the fan to stop it spinning — spinning from air can damage bearings)
- Clean the CPU heatsink fins — dust here causes overheating
- Clean the drive bay ventilation slots
- Wipe the outer shell with a dry cloth

7 Remove the HDD Bay Cage Screws

~4 min

The internal HDD bay cage is held in by 4 Phillips screws on the chassis, plus 1 screw on the I/O board. You must remove these to slide the motherboard out.

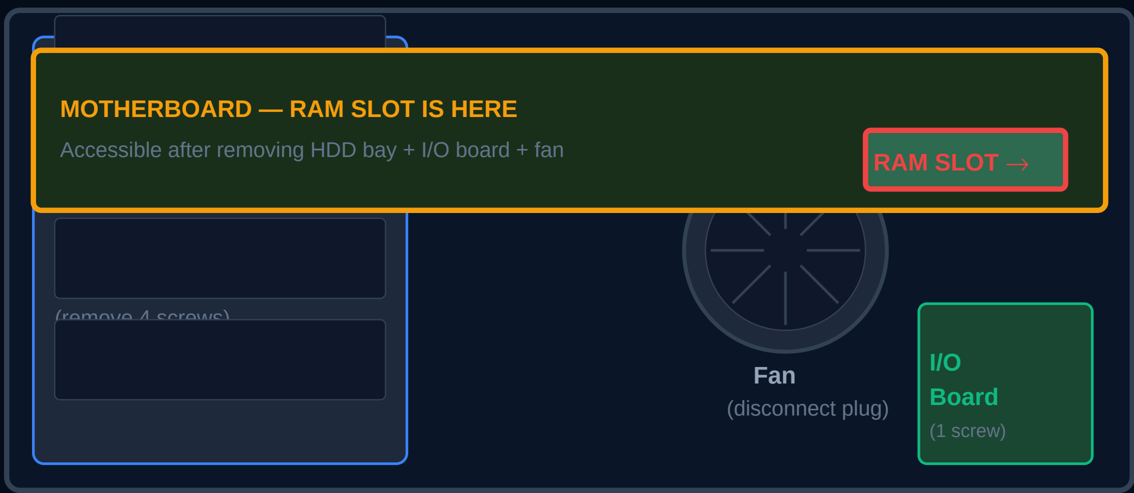

Figure 5: Internal layout (with case removed, viewed from top)

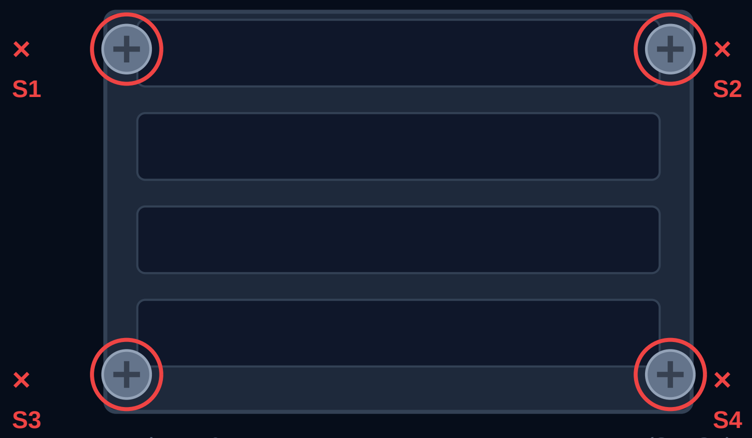

Figure 6: HDD cage — remove 4 corner screws (S1–S4)

- There are 4 screws securing the HDD bay cage to the chassis (one at each corner)
- There is 1 additional screw securing the I/O board (the rear port board)
- Remove all 5 screws — they are all the same type/size
- Keep them together in your small container

8 Disconnect the Fan and Remove Internals

~4 min

The fan is connected by a small 2-pin or 3-pin power header to the motherboard. You need to disconnect this to get the HDD cage fully out.

- Locate the fan power header — a small white or black plug near the fan
- Gently pull the header straight off its pin (do not yank the wires)
- You can now lift the HDD bay cage out of the chassis — lift from the front
- Remove the I/O board (rear port board) by lifting it upward from its connector
- The motherboard is now visible at the bottom of the chassis

Handle with care

The I/O board connects via a small ribbon or pin header. Lift it straight UP — do not tilt or bend. The fan cable is short — do not pull on the wires themselves, grip the plastic connector.

9 Remove the 4 Motherboard Screws

~3 min

With the HDD cage removed, the motherboard is visible. It is held down by 4 screws.

- Locate the 4 screws securing the motherboard to the chassis (one at each corner of the board)
- Remove all 4 screws
- Slide the motherboard out from the FRONT of the unit
- The board slides out on guides — start from the front edge and slide toward you

10 Remove the Old RAM Module

~3 min

The RAM slot is on the motherboard. With the board out, it is easily accessible.

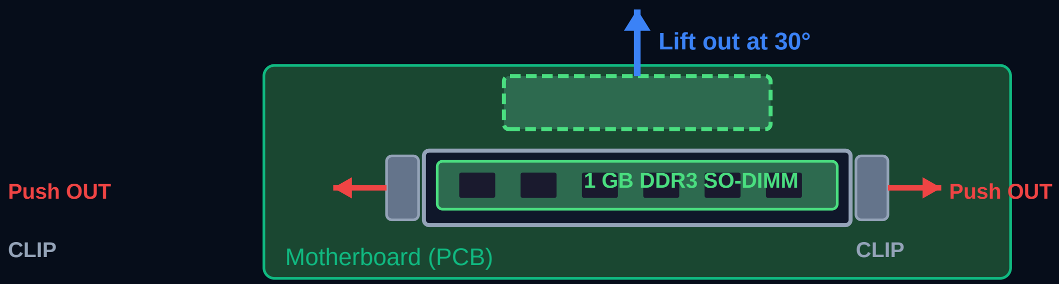

Figure 7: RAM slot — push retention clips outward, module pops up at ~30°, lift out

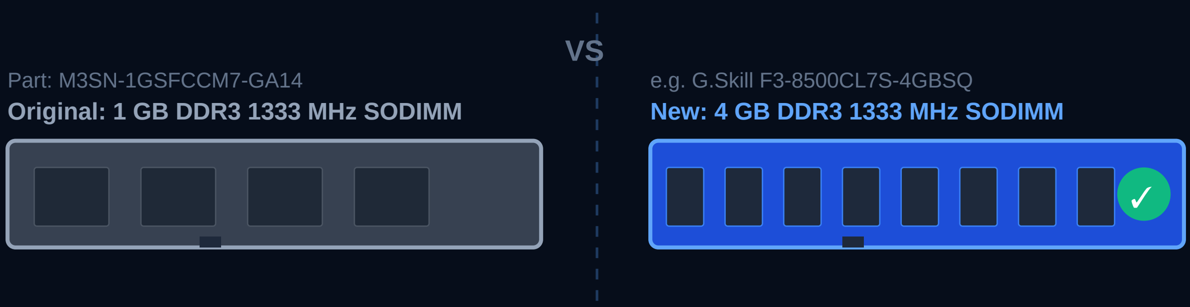

Figure 9: Old 1 GB module (left) vs new 4 GB module (right) — same connector type

- Locate the SO-DIMM slot — it holds a small green PCB (the 1 GB RAM stick)
- On each side of the slot is a small metal retention clip
- Use your thumbs to push BOTH clips simultaneously OUTWARD (away from the RAM module)
- The RAM module will spring up at approximately 30 degrees
- Grip the module by its edges (not the chips) and pull it out at that 30-degree angle
- Set the old module aside — do not throw it away until the upgrade is verified

11 Insert the New 4 GB RAM Module

~4 min

Figure 8: Insert new RAM at 30° into slot, press until clips snap closed with a click

■ Align the notch on the RAM module with the notch in the slot — it only fits ONE way

- Hold the new SO-DIMM module by its edges — never touch the gold contacts
- Look at the slot: there is a small notch (gap) that aligns with a notch on the module
- The RAM only fits ONE way — if it does not slide in easily at the angle, flip it 180°
- Insert the module into the slot at approximately 30 degrees
- With the module fully seated into the slot at 30°, press DOWN firmly on the module
- You will hear and feel a CLICK as the retention clips snap back into place
- If no click: the module is not fully seated — remove and try again
- Do NOT force it — if it requires heavy force, something is misaligned

✓ Seated correctly when:

Both retention clips are clipped over the notches on each end of the module. The module sits flat in the slot (not at an angle). You heard a click when pressing it down.

12 Reassemble in Reverse Order

~8 min

Reassembly is the exact reverse of disassembly. Follow this checklist:

- ■ Slide motherboard back into chassis from front — ensure it sits on the guide rails
- ■ Replace 4 motherboard screws — do not overtighten, just snug
- ■ Reconnect the I/O board — press straight down onto the connector
- ■ Place HDD bay cage back — align with guides and press into position
- ■ Reconnect the fan power header — press until it clicks
- ■ Replace 4 HDD cage screws and 1 I/O board screw (5 total)
- ■ Lower the outer shell — mind the top tabs — then slide FORWARD ~1 cm
- ■ Replace the 2 rear panel screws
- ■ Reinstall all 4 hard drive trays — slide in firmly until they click
- ■ Reconnect power and ethernet cables

■ Screw count check

Before powering on, verify: 2 rear screws + 1 I/O screw + 4 HDD cage screws + 4 motherboard screws = 11 screws total. If you have any screws left over, something is not re-assembled correctly.

13 Power On and Verify

~5 min

DSM — Control Panel — Info Center

Model Name:	DiskStation DS412+
Serial Number:	XXXXXXXXXX
DSM Version:	DSM 6.2.4-25556
Physical Memory:	4096 MB ← SUCCESS!
CPU Usage:	12%

Figure 10: DSM Control Panel → Info Center shows 4096 MB = upgrade successful

- Press the power button on the front of the DS412+
- The power LED will light blue, drives will spin up — normal startup takes 2–4 minutes
- Wait for the STATUS LED to turn solid green (blue during boot, green when ready)
- Open DSM in your browser: http://your-nas-ip:5000
- Log in with your admin account
- Go to: Control Panel → Info Center → General tab
- Confirm Physical Memory shows 4096 MB (or 2048 MB if you installed 2 GB)

Run the memory test (optional but recommended)

Synology provides a built-in memory test via the Synology Assistant desktop app:

- Download Synology Assistant from synology.com/en-global/support/download
- In the app, right-click your NAS → Memory Test
- The memory test runs for several hours — overnight is ideal
- A single pass with zero errors confirms the module is compatible and seated correctly
- If errors appear: power off, reseat the RAM, and retry

Verify from SSH

```
$ free -m                 total used free Mem: 3855 xyz xyz
```

Troubleshooting

NAS does not boot at all after upgrade

→ Power off immediately and remove the RAM module
→ Reseat the original 1 GB module and verify it boots — if yes, the new module is incompatible
→ Check the module part number matches a confirmed compatible module from the list
→ Check the module is 1.5V (not 1.35V low-voltage) — the D2700 requires standard 1.5V DDR3
→ Try pressing the module in more firmly — it must CLICK into both retention clips

NAS boots but shows 1 GB instead of 4 GB

→ The new module is not fully seated — power off and reseat it
→ Check both retention clips are locked over the module notches
→ The slot may not have released fully before inserting — check for bent pins in the slot

NAS is slow or freezes after upgrade

→ This is a known issue with some 4 GB modules on the DS412+
→ Try a different 4 GB module from the confirmed list (G.Skill F3-8500CL7S-4GBSQ is most reliable)
→ If the problem persists with any 4 GB module, use 2 GB instead — it runs rock-solid
→ Run the Synology memory test to confirm the module has no errors

Hard drives not detected after reassembly

→ Ensure each drive tray is fully seated — push in firmly until you feel it click
→ Check the SATA backplane connector inside — it may have been dislodged
→ Verify the fan power header is reconnected (the NAS may not fully start without it)

Outer shell will not go back on

→ The tabs must align with the metal bars on the left side before lowering
→ Slide the shell fully rearward first, then lower — the top tabs must pass over their hooks
→ If a tab was broken during removal, the shell will still fit snugly due to the rear screws

After Successful Upgrade — Next Step

With 4 GB RAM confirmed, your DS412+ is ready to run UAVLogBook via Docker. Return to the UAVLogBook Synology Docker Install Guide and follow the instructions starting from Section 2 (Install Docker from Package Center).